

FIONA JUSTINAMALADIRAVIAM PATRICIA JOSELYN

Passionate VLSI graduate student with 3 years of strong combined academic and hands-on industry experience in ASIC/FPGA design using verilog, verification and timing closure seeking HW summer internships.

fjustina@purdue.edu | +1 765 746 9680 | Locations: Across USA | Languages: English, Tamil | [LinkedLn](#)

INTERESTS

RTL design | STA | Digital design |

EDUCATION

Masters in Electrical and Computer Engineering

Specializing in VLSI and circuit design

Purdue university

Relevant coursework

- MOS VLSI with project using Cadence virtuoso
- Computer architecture with project using Gem5

Bachelor of Engineering in Electronics and Communication Engineering

Sri Venkateswara College of Engineering(SVCE) CGPA - 9.58/10

Aug 2025 - May 2027

West Lafayette, Indiana,
USA

Aug 2018 - May 2022

Chennai, India

PROFESSIONAL EXPERIENCE

MBIT Wireless Pvt Ltd

RTL Design Engineer

Aug 2022 - May 2025

Chennai, India

- Engineered and optimized signal processing algorithms in Verilog for a custom 4G modem IC achieving **low power and low area** implementation using **Cadence Genus and Synopsys** tools for analysis in 65nm.
- Derived and validated ASIC and FPGA timing constraints from scratch for both full chip and 3 subsystems for 2 projects through in-depth architectural analysis in **Xilinx and cadence genus** by using TCL.
- Proposed architectural modifications and implemented targeted slack fixes via refined constraint, achieving timing closure improvements to **50-100ps**.
- Developed and executed new 50+ **design-oriented verification scenarios** and developed 10+ standalone TBs from scratch for robust verification with functional coverage of 92%.

PROJECTS

Goldschmidt algorithm based IEEE 32 bit floating point divider

Guided by Dr. Athappan, Professor, SVCE

- Implemented the divider algorithm entirely from scratch in Verilog using EDA tools from research papers
- Designed 9 different arithmetic units from scratch and replaced in the divider implementation.
- Performance analysis using Cadence Genus and observed ~7% decrease in delay than the preexisting design

Rural water distribution refurbishment with secured payment gateway

Guided by Mrs.R.Kousalya, Professor, SVCE

- Developed a new project on 95% automation of rural water distribution using indigenous Shakti/Vega processors.
- Authored and presented a paper to the smart India hackathon council as a lead for a team of 5.

Wireless power transmission to Micro UAVs using FPGA control algorithms

Guided by Dr.Saeed Mohammadi, Professor, Purdue university

- Ongoing project on WPT using RF and communication through FPGA targeting 60% power reception efficiency

SKILLS

Languages: Verilog HDL, TCL, Python, System Verilog

Skills: Low power/area design, STA, scenario listing, coverage, Microarchitecture, Linux, Lint, Schematic/ layout design

Tools: Synopsys VCS, Cadence - genus, Innovus, Virtuoso, Gem5, Xilinx

ACHIEVEMENTS

- **Awards/Activities:** **Gold medalist B.E-ECE**(In a class of 180)
Best Outgoing ECE student award

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EXPERIENCES

Social Media Coordinator-Institution Innovation council

- Conducting Innovations and patenting related workshops for 200+ sophomores, juniors and seniors across departments and monitored the social interactions and advertisements for IIC.

Executive member -ECEA-SVCE

- Marketing lead of the national level technical symposium of the ECE department.
- Conducted 100+ events including webinars, seminars and technical competitions for 3000+ students in India.